

01 OPTIMIZATION

Gradient descent

w_{t+1} = w_t - \eta_t \nabla f(w_t).

For quadratic f(w) = 1/2 w^T H w - b^T w, H > 0, let L = \lambda_{max}(H).

COND 0 < \eta < 2/L gives convergence to the unique minimizer. \eta = 1/L is a safe choice. ! For this quadratic, 0 < \eta < 2/L also decreases the objective; 1/L is a common sufficient bound for general smooth objectives. EX f(w) = 2w^2: L = 4; w_{t+1} = (1 - 4\eta)w_t. At \eta = 1/2, oscillation persists.

Convexity

f(y) \ge f(x) + \nabla f(x)^T (y - x).

\nabla^2 f \ge 0 implies convexity on a convex domain; \nabla^2 f \ge \mu I, \mu > 0, implies strong convexity. COND Differentiable convex f: \nabla f(w) = 0 implies a global minimum.

! Stationary point alone is insufficient for a nonconvex objective.

Newton / stochastic update

w^+ = w - [\nabla^2 f(w)]^{-1} \nabla f(w).

Solve Hp = g then set w^+ = w - p; avoid forming the inverse. SGD uses a sample or minibatch gradient. ! Newton needs a suitable Hessian and often damping; SGD gradients fluctuate.

02 DERIVATIVE TOOLKIT

For column vectors and symmetric A:

\nabla_w (a^T w) = a,

\nabla_w (w^T A w) = 2A w,

\nabla_w \frac{1}{2} \|Xw - y\|^2 = X^T (Xw - y).

If A is not symmetric, gradient is (A + A^T)w.

Chain rule z = Ax + b, L = L(z):

\nabla_x L = A^T \nabla_z L.

COND Check dimensions: X \in \mathbb{R}^{n \times d}, w \in \mathbb{R}^d, y \in \mathbb{R}^n. EX z = 3w + 1, L = z^2: dL/dw = 6z. ! A sum loss and a mean loss differ by n in the gradient and regularization scale.

03 CONSTRAINED OPTIMIZATION

For min f(w) subject to g_i(w) \le 0 and h_j(w) = 0:

\mathcal{L} = f + \sum_i \alpha_i g_i + \sum_j \nu_j h_j.

KKT checklist Stationarity \nabla_w \mathcal{L} = 0; primal feasibility; dual feasibility \alpha_i \ge 0; complementary slackness \alpha_i g_i = 0. COND Convex problem + suitable constraint qualification (e.g. Slater) gives the usual optimality/strong-duality result. ! Active constraint does not force a strictly positive multiplier; \alpha_i > 0 does force activity. EX min x subject to x \ge 1: g = 1 - x, x^* = 1, \alpha^* = 2.

04 PROXIMAL / NUMERICAL CHECKS

Soft threshold For min_u \frac{1}{2} \|u - v\|^2 + \tau \|u\|_1, u_j = \text{sign}(v_j) \max(|v_j| - \tau, 0).

COND Proximal gradient for g(w) + \lambda \|w\|_1 uses v = w - \eta \nabla g(w) and \tau = \eta \lambda. EX v = (2, -4), \tau = .5: u = (1.5, 0). ! Threshold depends on the step size; do not use \lambda alone.

Gradient check For a coordinate direction e_j: g_j \approx \frac{f(w + he_j) - f(w - he_j)}{2h}.

Compare analytical and numerical gradients away from nondifferentiable points; use double precision and several step sizes. ! Too small h amplifies rounding error; too large h gives truncation error. Stopping rule Monitor objective, gradient norm, and relative parameter change. Report tolerance and maximum iterations. COND Small steps alone can reflect a small learning rate rather than proximity to an optimum.

05 LINEAR REGRESSION

J(w) = \frac{1}{2n} \|Xw - y\|^2.

\nabla J = \frac{1}{n} X^T (Xw - y).

Normal equation X^T X w = X^T y. Full column rank gives a unique least-squares solution. Otherwise use a pseudoinverse or regularization. COND An intercept is a column of ones; check whether it is penalized. EX X = (1, 2)^T, y = (2, 4)^T: w^* = 2. ! In code, use least-squares/QR/SVD solvers instead of computing (X^T X)^{-1}. Residuals r = y - X\hat{w}; at an unregularized least-squares optimum, X^T r = 0.

06 REGULARIZATION

Ridge

J = \frac{1}{2n} \|Xw - y\|^2 + \frac{\lambda}{2} \|w\|^2.

(X^T X + n\lambda I)w = X^T y.

COND \lambda > 0 makes this matrix positive definite. Standardize features before comparing penalty strength. ! If data loss has no 1/n, the normal equation has \lambda I under the matching 1/2 convention. Lasso J = \frac{1}{2n} \|Xw - y\|^2 + \lambda \|w\|_1. For w_j = 0, the subgradient is [-1, 1]; lasso can set coefficients exactly to zero.

Bias / variance For squared prediction error at fixed x:

\mathbb{E}[(Y - \hat{f}(x))^2] = \sigma^2 + \text{Bias}^2 + \text{Var}(\hat{f}(x)).

COND Expectation is over training sets and independent test noise; \hat{f}(x) = \mathbb{E}[Y | x]. ! Stronger regularization often raises bias and lowers variance; this is a tendency, not a universal monotonic law.

07 LOGISTIC REGRESSION

y \in \{0, 1\}; z = w^T x + b:

p = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad \sigma' = p(1 - p).

\ell = -y \log p - (1 - y) \log(1 - p).

\partial \ell / \partial z = p - y, \quad \nabla_w \ell = (p - y)x.

Stable logit loss

\ell = \max(z, 0) - yz + \log(1 + e^{-|z|}).

COND Decision threshold p \ge \tau is z \ge \log \frac{\tau}{1-\tau} for 0 < \tau < 1. EX z = 0, y = 1: p = 0.5, logit gradient = -0.5. ! Logits are unrestricted real numbers; probabilities lie in [0, 1]. Threshold 0.5 is a choice. Multiclass p_k = e^{z_k} / \sum_j e^{z_j}, \ell = -\sum_k y_k \log p_k; \partial \ell / \partial z_k = p_k - y_k for a target distribution with \sum_k y_k = 1. ! Subtract \max_j z_j before exponentiating.

08 LINEAR MODEL DECISIONS

Perceptron If y_i(w^T x_i + b) \le 0, update

w \leftarrow w + \eta y_i x_i, \quad b \leftarrow b + \eta y_i.

COND Labels \{-1, +1\}. Classical finite-mistake convergence needs separability and bounded inputs. ! A perceptron does not generally maximize the margin or produce calibrated probabilities. MAP interpretation A Gaussian prior gives an L_2 penalty; a Laplace prior gives an L_1 penalty, with scale matched to the likelihood convention. Feature transforms Polynomial features allow nonlinear boundaries with a model still linear in its parameters. Add interactions only using training/validation evidence. EX Features (1, x, x^2): prediction w_0 + w_1 x + w_2 x^2. ! Linear in parameters does not mean linear in the original input. Cost-sensitive threshold For calibrated p = P(y = 1 | x) and only false-positive/false-negative costs: \tau = \frac{C_{FP}}{C_{FP} + C_{FN}}. COND Both costs nonnegative with positive sum; predict positive when p \ge \tau. ! A changed class prior can invalidate probability calibration and the chosen threshold.

09 SVM / MARGINS

y_i \in \{-1, +1\}; f_i = w^T x_i + b.

\gamma_i = \frac{y_i f_i}{\|w\|}.

Hard margin

\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{s.t. } y_i f_i \ge 1.

Canonical margin width is 2/\|w\|.

COND Hard margin requires linear separability in the chosen feature space. ! Positive rescaling of (w, b) changes functional margin, while geometric margin is unchanged. Soft margin

\min \frac{1}{2} \|w\|^2 + C \sum_i \xi_i, \quad \xi_i \ge 0,

y_i f_i \ge 1 - \xi_i. Equivalently use hinge loss \max(0, 1 - y_i f_i). Dual / KKT 0 \le \alpha_i \le C, \sum_i \alpha_i y_i = 0; w = \sum_i \alpha_i y_i \phi(x_i). COND 0 < \alpha_i < C \Rightarrow y_i f_i = 1; \alpha_i = C allows points on/inside the margin. ! \alpha_i = C does not imply misclassification; \alpha_i = 0 can occur on the margin in degenerate cases. EX y_i f_i = 0.3: hinge loss = 0.7; prediction has the correct sign but lies inside the margin.

10 KERNELS

K(x, z) = \phi(x)^T \phi(z).

Validity Symmetric kernel; every finite Gram matrix G_{ij} = K(x_i, x_j) must be positive semidefinite: c^T G c \ge 0 for all c.

Common forms Linear x^T z; polynomial (x^T z + c)^p for integer p \ge 1, c \ge 0; RBF e^{-\gamma \|x - z\|^2}, \gamma > 0. COND Nonnegative sums and pointwise products of PSD kernels remain PSD. ! A symmetric similarity function alone need not be a valid kernel. EX G = [\frac{1}{2} \frac{1}{1}] has eigenvalue -1: fails PSD. Prediction f(x) = \sum_i \alpha_i y_i K(x_i, x) + b.

Only nonzero coefficients contribute; no explicit \phi is needed.

11 DISTANCE / CLUSTERING

k-means

\min_{\mu,a} \sum_i \|x_i - \mu_{a_i}\|^2.

Alternate nearest-centroid assignment and mean updates; objective does not increase in either exact step. COND Squared Euclidean distance; fix the treatment of empty clusters. Use multiple initializations. ! Convergence is to a local solution; cluster IDs have no intrinsic order. EX Points 0, 2, 9, 11, k = 2: groups \{0, 2\}, \{9, 11\} have means 1, 10. Cosine similarity s(x, z) = \frac{x^T z}{\|x\| \|z\|}. COND Both vectors nonzero. Normalize first when direction rather than magnitude is relevant. ! Cosine similarity is not Euclidean distance; a zero vector needs an explicit convention.

16 SVM DUAL / WORKED CHECK

\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j), \quad 0 \le \alpha_i \le C, \quad \sum_i \alpha_i y_i = 0.

COND This form assumes an unpenalized bias. Recover b = y_i - \sum_j \alpha_j y_j K(x_j, x_i) from any 0 < \alpha_i < C. EX Linear hard-margin sample: (x_1, y_1) = (-1, -1), (x_2, y_2) = (1, +1). w = 1, b = 0, \alpha_1 = \alpha_2 = 1/2; margin width = 2. ! If no coefficient is strictly between 0 and C, a free support-vector formula cannot be used directly; use KKT bounds for b.

12 DECISION TREES

Class fractions p_k in a node:

G = 1 - \sum_k p_k^2, \quad H = -\sum_k p_k \log p_k.

Use 0 log 0 = 0. For a split into L, R:

\Delta I = I(P) - \frac{n_L}{n} I(L) - \frac{n_R}{n} I(R).

COND Choose the split maximizing impurity reduction; for regression, minimize weighted squared error. EX Binary node (0.5, 0.5): Gini = 0.5; pure children give gain 0.5. ! Weight child impurities by their sample counts; averaging them equally is generally wrong. Capacity control Depth limit, minimum leaf size, pruning. Deep trees fit interactions but may have high variance. COND Axis-aligned splits do not require feature standardization.

13 BAGGING / RANDOM FORESTS

Train models on bootstrap samples; average regression predictions or vote for classification.

\text{Var}(\bar{h}) = \sigma^2 \left(\rho + \frac{1 - \rho}{B} \right).

COND Formula assumes equal individual variance and pairwise correlation \rho. Random forests additionally sample candidate features at each split to reduce correlation. Out-of-bag Evaluate each observation using only trees whose bootstrap samples excluded it. ! Many highly correlated trees retain a variance floor; B alone cannot remove \rho \sigma^2. EX \rho = 0, B = 10: variance \sigma^2/10; \rho = 0.5: 0.55 \sigma^2.

14 BOOSTING

Gradient boosting

r_{im} = - \left. \frac{\partial \ell(y_i, F)}{\partial F} \right|_{F=F_{m-1}(x_i)}.

Fit h_m to pseudo-residuals; update F_m = F_{m-1} + \eta \gamma_m h_m. EX Squared loss \ell = \frac{1}{2} (y - F)^2: pseudo-residual = y - F. AdaBoost \epsilon_m = \sum_i q_i 1[y_i \neq h_m(x_i)]; a_m = \frac{1}{2} \log \frac{1 - \epsilon_m}{\epsilon_m}; q_i^+ \propto q_i e^{-a_m y_i h_m(x_i)}. COND Binary h_m, y_i \in \{-1, +1\}; the standard positive-weight step needs 0 < \epsilon_m < 1/2. ! Boosting fits models sequentially. Noisy labels can receive large weight.

15 ENSEMBLE DECISIONS

Aggregation Regression: mean of predictions. Classification: vote or average aligned class probabilities, then apply the decision rule. COND Align class order before averaging probabilities from different models. ! Averaging logits and averaging probabilities give different ensembles. Stacking Train a second-level model on out-of-fold base-model predictions; refit base models on full training data for deployment. COND All folds and feature processing obey the same leakage constraints as ordinary CV. ! Training the stacker on in-sample predictions gives it an overly optimistic view of base models. EX Two binary probabilities .2, .8: probability average = .5. Agreement and confidence remain separate questions. Choose / check Tune tree capacity and ensemble size with validation; record metric, split, seed, and hyperparameters.

17 PROBABILITY / BAYES

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}, \quad P(B) > 0.$$

Total probability For a disjoint, exhaustive partition $\{A_k\}$:

$$P(B) = \sum_k P(B \mid A_k)P(A_k).$$

EX Prevalence 0.01, sensitivity 0.9, false-positive rate 0.05:

$$P(D \mid +) = \frac{.9(.01)}{.9(.01) + .05(.99)} \approx .154.$$

! Sensitivity $P(+ \mid D)$ differs from the posterior $P(D \mid +)$. Include false positives in the denominator.

Expectation / variance
 $E[aX + b] = aE[X] + b$;
 $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
 $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$.

COND Independence implies zero covariance when moments exist. The converse generally fails.

18 CLASSIFICATION METRICS

$$\text{Prec} = \frac{TP}{TP + FP}, \quad \text{Rec} = \frac{TP}{TP + FN}.$$
$$F_1 = \frac{2TP}{2TP + FP + FN}.$$

Specificity = $TN / (TN + FP)$; accuracy = $(TP + TN) / N$.

COND Specify the positive class and handle zero denominators explicitly.

! Accuracy can hide failure on a rare class. Macro averaging weights classes equally; micro pooling weights counts.

EX $TP = 8, FP = 2, FN = 4$: precision = .8, recall = $2/3, F_1 = 8/11$.

Thresholds Raising a score threshold cannot increase predicted positives; recall is nonincreasing. Precision need not increase monotonically.

! ROC uses $FPR = FP / (FP + TN)$; a precision-recall curve uses precision.

19 VALIDATION / LEAKAGE

Procedure Split data $\rightarrow$ fit transforms on training data $\rightarrow$ tune on validation/CV $\rightarrow$ evaluate once on held-out test data.

COND Refit preprocessing inside every CV fold. Split by time, person, or group when the task requires it.

! Scaling, feature selection, imputation, and target encoding can all leak held-out information.

Model selection Validation chooses hyperparameters. Test estimates generalization after choices are fixed. Nested CV separates inner tuning from outer evaluation.

EX Repeated patients: keep all records of a patient in the same fold.

! Picking a threshold from test labels is also test-set tuning.

Distance-based models Standardize features when units differ; k -NN stores training data, and prediction depends on distance and k .

COND Fit centering/scaling statistics on the training split only.

20 ESTIMATION / UNCERTAINTY

Sample mean Independent identically distributed X_i with finite variance σ^2 :

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \sigma^2 / n.$$

Sample variance $s^2 = \sum_i (X_i - \bar{X})^2 / (n - 1)$ is unbiased for $n > 1$.

COND The $1/n$ variance result depends on independence; correlated observations add covariance terms.

Gaussian mean interval Known σ : $\bar{X} \pm z_{1-\alpha/2} \sigma / \sqrt{n}$; unknown σ : use t_{n-1} and s for normal samples.

! A frequentist confidence level describes repeated-sampling coverage.

EX Known $\sigma = 2, n = 100$: standard error = .2, approximate 95% half-width = .392.

Bootstrap Resample observations with replacement; recompute the statistic. Respect groups or time dependence with an appropriate resampling scheme.

! A narrow interval cannot repair selection bias or a mismatched evaluation population.

21 NEURAL NETS / BACKPROP

Layer $a_\ell = W_\ell h_{\ell-1} + b_\ell, h_\ell = \phi(a_\ell)$.
 $\delta_\ell = (W_{\ell+1}^T \delta_{\ell+1}) \odot \phi'(a_\ell)$.
 $\nabla_{W_\ell} L = \delta_\ell h_{\ell-1}^T, \quad \nabla_{b_\ell} L = \delta_\ell$.

COND These expressions are per example with column activations. Sum/average consistently across a batch.

Activations ReLU $\max(0, a)$: derivative 1 for $a > 0, 0$ for $a < 0$; at 0 choose a subgradient. Sigmoid derivative $p(1 - p)$; tanh derivative $1 - h^2$.

! Apply the activation derivative elementwise; transpose the downstream weight matrix.

EX $h = \text{ReLU}(wx), x = 2, w = 1, L = \frac{1}{2}(h - 3)^2$: $dL/dw = -2$.

Training / inference Dropout is stochastic during training; batch normalization uses batch statistics during training and stored statistics at inference.

! Switch to evaluation mode for inference.

22 PCA / PROJECTIONS

Center $X \in \mathbb{R}^{n \times d}$ first. Let $X = U\Sigma V^T$ be an SVD.

$$Z = XV_k, \quad \hat{X} = ZV_k^T.$$

Top right singular vectors maximize projected variance and minimize rank- k squared reconstruction error.

Explained variance ratio
$$\text{EVR}_k = \frac{\sum_{j=1}^k \sigma_j^2}{\sum_j \sigma_j^2}.$$

COND Sample covariance is $X^T X / (n - 1)$ for centered observations. Scale first if units should receive comparable influence.

! PCA is unsupervised. High variance need not predict the label; component signs are arbitrary. **EX** $X^T X = \text{diag}(9, 1)$: first component along axis 1, $\text{EVR} = .9$.

23 NAIVE BAYES / LIKELIHOOD

$$\hat{y} = \arg \max_c \left[\log P(c) + \sum_j \log P(x_j \mid c) \right].$$

COND Assumes features are conditionally independent given the class. Choose a likelihood matching feature type. **Multinomial smoothing**

$$P(j \mid c) = \frac{N_{cj} + \alpha}{\sum_r N_{cr} + \alpha V}.$$

V is vocabulary size; $\alpha > 0$ avoids zero token probabilities.

! Conditional independence is different from unconditional independence. Accumulate logs to avoid underflow.

EX One unseen token makes an unsmoothed product zero; additive smoothing preserves a finite log score.

24 MODEL / LOSS MATCHING

Regression losses Squared error targets a conditional mean; absolute error targets a conditional median, when the relevant expectations exist.

Classification losses Cross-entropy trains a probability model; hinge loss targets a margin. Their numerical values are not directly comparable.

COND Compare models on the same held-out metric and data split.

Learning rate / batch size Larger batches reduce sampling noise under ordinary independent sampling. Learning rate, batch size, and schedule interact.

! More epochs need not improve held-out performance; select early stopping using validation data.

EX Train loss falls while validation loss rises: inspect capacity, regularization, leakage, and split mismatch.

Reproducibility Record preprocessing, seed, split identities, metric convention, and model settings; preserve a held-out test set.

! A fixed seed is not evidence that evaluation data represent deployment data.

25 ASYMPTOTIC ANALYSIS

Bounds $f = O(g)$: eventual upper bound; $f = \Omega(g)$: lower bound; $f = \Theta(g)$: both. State what n measures.

Common growth
 $1 < \log n < n < n \log n < n^2 < 2^n < n!$ asymptotically.

Loop patterns Halving/doubling a positive index: $\Theta(\log n)$. Triangular loop $\sum_{i=1}^n i$: $\Theta(n^2)$. **EX** Outer loop doubles i ; inner loop runs i times: $1 + 2 + 4 + \dots \leq 2n$, hence $\Theta(n)$.

! Multiply costs only when inner work has the claimed cost on every outer iteration.

Master theorem For $T(n) = aT(n/b) + f(n)$, $a \geq 1, b > 1$, compare f to $n^{\log_b a}$. Polynomially smaller: $\Theta(n^{\log_b a})$; equal: $\Theta(n^{\log_b a} \log n)$; polynomially larger plus regularity: $\Theta(f(n))$.

COND The third case needs $af(n/b) \leq cf(n)$ eventually for some $c < 1$.

! These three basic cases do not cover every recurrence.

26 SEARCH / SORT

Binary search invariant Search $[l, r)$; $m = l + (r - l)/2$. Lower bound finds the first $a[m] \geq x$.

- If $a[m] < x$, set $l = m + 1$.
- Otherwise set $r = m$; stop at $l = r$.

COND Array sorted in the comparison order. Result may equal n .

! For membership, also check $l < n$ and $a[l] = x$.

Sort comparison

Method	Time	Stable?
Merge	$\Theta(n \log n)$	yes*
Heap	$O(n \log n)$	no
Quick	avg. $O(n \log n)$	no
Insertion	worst $O(n^2)$	yes*

COND *Standard tie-preserving implementation. Quicksort worst case is $O(n^2)$. **EX** Lower bound of 2 in $[1, 2, 4]$ is index 1, with zero-based indexing.

27 DATA STRUCTURES

Choose by operation Stack: LIFO; queue: FIFO; hash map: expected constant-time lookup under suitable hashing/load; balanced BST: $O(\log n)$ lookup/update.

Binary heap Peek $O(1)$; insert/extract $O(\log n)$; bottom-up heap construction $O(n)$.

! A heap is partially ordered; arbitrary-value search can take $O(n)$.

Amortized cost Dynamic array doubling gives amortized $O(1)$ append, although a resizing append costs $O(n)$.

EX n pushes with doubling copy $1 + 2 + 4 + \dots < n$ old elements in total.

! Expected, amortized, and worst-case bounds describe different guarantees.

28 UNION-FIND / PREFIX SUMS

Disjoint sets $\text{find}(x)$ returns a representative; $\text{union}(x, y)$ merges sets. Use path compression plus union by size/rank.

COND A sequence of operations has near-constant amortized cost $O(\alpha(n))$ per operation under the standard implementation.

! The representative is an implementation choice; it is not necessarily the smallest member.

EX Union $(0, 1)$ and $(1, 2)$ makes $\text{find}(0) = \text{find}(2)$ true.

Prefix sum Define $P[0] = 0, P[i + 1] = P[i] + a[i]$.

$$\sum_{i=l}^{r-1} a[i] = P[r] - P[l].$$

COND Half-open range $[l, r)$; preprocessing $O(n)$, query $O(1)$ on an unchanged array.

! Use a numeric type large enough for accumulated sums. Point updates invalidate later prefix values.

EX $a = [2, 3, 5], P = [0, 2, 5, 10]$; range $[1, 3)$ has sum $10 - 2 = 8$.

29 GRAPH TRAVERSAL

Let V be vertex count and E edge count; use adjacency lists.

BFS Queue; mark a vertex discovered when enqueueing it. Finds minimum edge-count paths from a source in $O(V + E)$.

COND Unweighted or equal-weight edges for shortest-path interpretation.

! Marking only on dequeue may enqueue the same vertex repeatedly.

DFS Recursion/stack; discovery/finish states. Directed cycle exists when an edge points to a vertex on the active DFS stack.

COND In an undirected simple graph, ignore the parent edge when checking for cycles.

EX $s \rightarrow a, s \rightarrow b, a \rightarrow t$: BFS gives distances $(0, 1, 1, 2)$.

Topological ordering Kahn: enqueue zero-indegree vertices, remove them and decrement neighbor indegrees. Fewer than V outputs implies a directed cycle.

! A topological ordering exists exactly for a directed acyclic graph.

30 SHORTEST PATHS / MST

Relaxation
$$d[v] \leftarrow \min(d[v], d[u] + w(u, v)).$$

Initialize source to 0, others to ∞ . **Dijkstra** Extract smallest tentative distance; relax outgoing edges. Binary heap:

$O((V + E) \log V)$ with decrease-key. **COND** All edge weights nonnegative. A lazy heap variant skips stale entries and has an E -dependent heap bound.

! Negative edges can invalidate settling a vertex. Do not add weights to an unreachable/infinite distance.

Bellman-Ford / DAG Bellman-Ford: $V - 1$ passes, $O(VE)$; a further relaxation detects a reachable negative cycle. DAG: topological relaxation, $O(V + E)$, negative edges allowed.

Minimum spanning tree Kruskal: sort edges, union endpoints from different components, $O(E \log E)$. Prim: grow a tree by the cheapest crossing edge.

COND Undirected weighted graph; disconnected input gives a minimum spanning forest.

! An MST minimizes total tree weight; it need not preserve shortest source-to-vertex paths.

31 DYNAMIC PROGRAMMING

Design steps State meaning $\rightarrow$ base cases $\rightarrow$ transition $\rightarrow$ dependency order $\rightarrow$ answer location. Time $\approx$ states $\times$ work/state; memory depends on stored dependencies.

0/1 knapsack
 $D[i, c] = \max\{D[i - 1, c], v_i + D[i - 1, c - w_i]\}.$

Second term only when $c \geq w_i$; base $D[0, c] = 0$ for optional selection and nonnegative capacity.

COND Each item is usable at most once. One-array implementation updates capacity downward.

! Ascending capacity reuses the current item, producing the unbounded version.

EX One item $(w, v) = (2, 3)$, capacity 4: 0/1 answer = 3; unbounded answer = 6.

Reconstruction Store choices or compare predecessor states. Rolling arrays save space but may remove the trace needed to reconstruct a solution.

32 DP / CODE MICRO-EXAMPLE

0/1 knapsack, one array Initialize $\text{dp}[0 \dots C]$ to zero, then:

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for (w, v) in items:
    for c = C down to w:
        dp[c] = max(dp[c],
                    v + dp[c-w])
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COND Integer nonnegative capacities; optional item selection. Time $O(nC)$, memory $O(C)$.

! $O(nC)$ is pseudopolynomial because capacity is encoded with $O(\log C)$ bits.

Longest common subsequence Matching characters: $D[i, j] = 1 + D[i - 1, j - 1]$; otherwise

$D[i, j] = \max(D[i - 1, j], D[i, j - 1])$. **COND** Base row/column zero; time $O(nm)$. A subsequence may skip positions.

! Longest common substring uses a different recurrence and requires contiguous matches.

EX ABC and AC: LCS length 2, common substring length 1.